Pratham Patel

+91 9016239929 | caffeinecompiler@duck.com | linkedin.com/in/realpratham | github.com/sadisticbrew

Education

National Institute of Engineering — Mysore, India
Bachelor of Engineering in Computer Science — *Sep. 2025 – present*

Awards

1st Place Winner | C Programming Bootcamp Hackathon (NIE Mysore) | Dec. 2025

Projects

Muon | *Go, C, eBPF, cilium/ebpf, Bubbletea* — GitHub

- Built a Linux process tracer using eBPF with a C kernel probe and Go user-space consumer, monitoring syscall events across entire process trees with kernel-space PID filtering via eBPF hashmaps.
- Benchmarked at <1% overhead on file and memory workloads and 12.7% on exec-heavy loads, vs. perf trace (67.5%) and strace (144%); validated across 7 runs with CPU frequency locked and cores isolated.
- Implemented CO-RE kernel probes via vmlinux.h with a 64MB BPF ring buffer, eliminating event drops during high-frequency syscall bursts.
- Built a Bubbletea TUI for real-time tracing output; ships as a zero-dependency single binary that attaches to any process by PID in under one second.

Horcrux | *Go, Cryptography, Google Tink* — GitHub

- Built a CLI tool for encrypting sensitive files using AES-256-GCM envelope encryption.
- Implemented a Shamir's Secret Sharing engine from scratch to split master keys into N shares, requiring M to reconstruct – no single share reveals anything about the key.
- Used Google Tink for constant-memory streaming of large files, avoiding full in-memory loads.
- Designed a custom binary metadata header (HRX2) encoding key shards and file metadata for deterministic reconstruction.

Stanza | *Python, Language Design* — GitHub

- Built a custom interpreted language from scratch in Python: lexer, recursive descent parser, and AST-based tree-walk interpreter supporting dynamic variables, first-class functions, and complex control flow.
- Wrote an error reporter that underlines the exact failing token in source, with line and column context.
- Runs in a managed execution environment with custom boolean types and strict type-checking, abstracting memory management from the user.

Technical Skills

Languages: Go, Python, Rust, C, C++, Bash
Developer Tools: Linux, Git, Vim/Zed, Cobra CLI, cilium/ebpf, Bubbletea
Core Focus: Systems Architecture, Compiler Design, Cryptography, Kernel Tracing